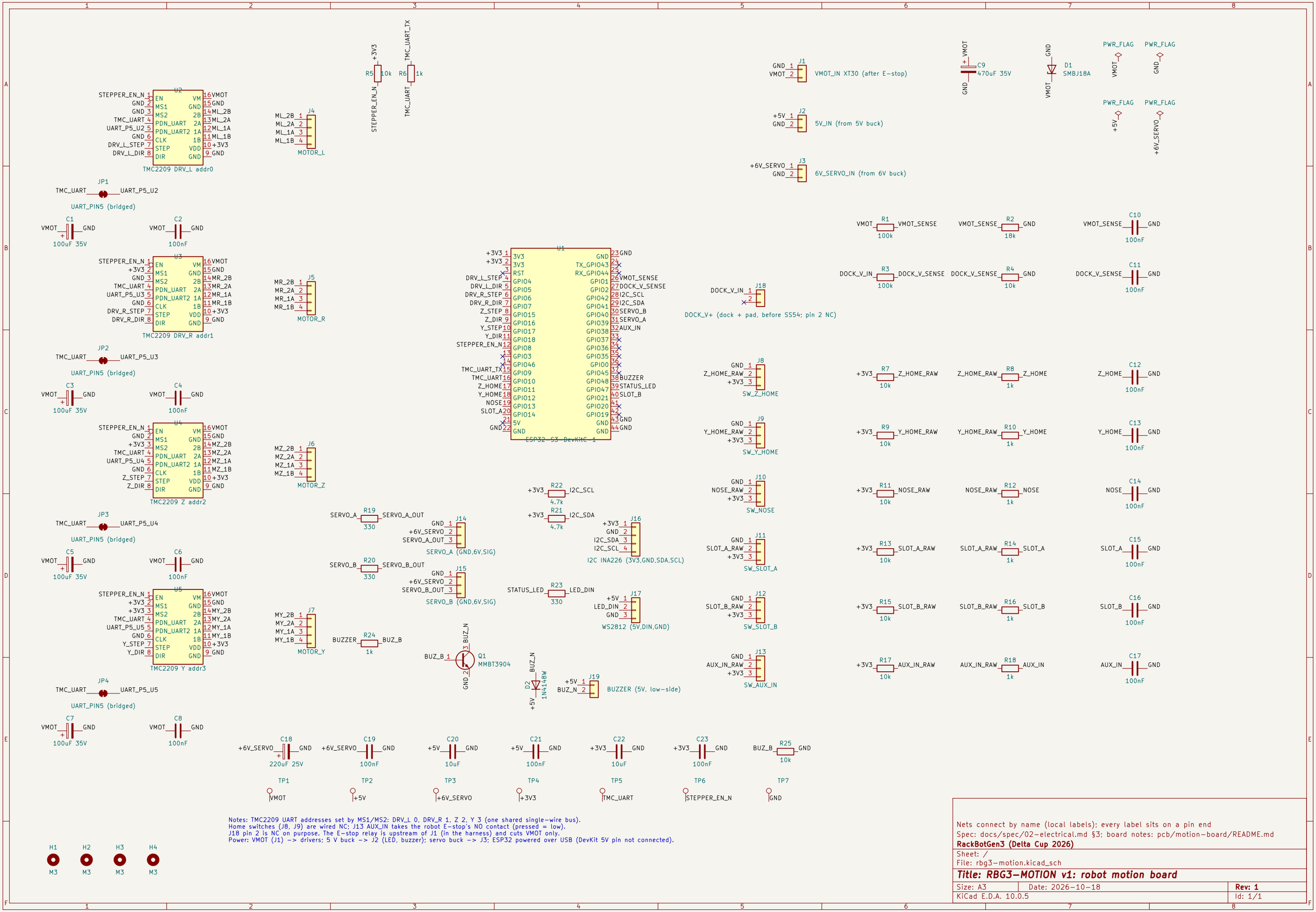

U2
TMC2209 DRV_L addr0
U3
TMC2209 DRV_R addr1
U4
TMC2209 Z addr2
U5
TMC2209 Y addr3
U1
ESP32-S3-DevKitC-1
J1
VMOT_IN XT30 (after E-stop)
J2
5V_IN (from 5V buck)
J3
6V_SERVO_IN (from 6V buck)
J4
MOTOR_L
J5
MOTOR_R
J6
MOTOR_Z
J7
MOTOR_Y
J8
SW_Z_HOME
J9
SW_Y_HOME
J10
SW_NOSE
J11
SW_SLOT_A
J12
SW_SLOT_B
J13
SW_AUX_IN
J14
SERVO_A (GND,6V,SIG)
J15
SERVO_B (GND,6V,SIG)
J16
I2C INA226 (3V3,GND,SDA,SCL)
J17
WS2812 (5V,DIN,GND)
J18
DOCK_V+ (dock + pad, before SS54; pin 2 NC)
J19
BUZZER (5V, low-side)
JP1
JP2
JP3
JP4
UART_PIN5 (bridged)
Q1
MMBT3904
D1
SMBJ18A
D2
1N4148W
C9
470uF 35V
PWR_FLAG
Notes: TMC2209 UART addresses set by MS1/MS2: DRV_L 0, DRV_R 1, Z 2, Y 3 (one shared single-wire bus).
Home switches (J8, J9) are wired NC; J13 AUX_IN takes the robot E-stop's NO contact (pressed = low).
J18 pin 2 is NC on purpose. The E-stop relay is upstream of J1 (in the harness) and cuts VMOT only.
Power: VMOT (J1) -> drivers; 5 V buck -> J2 (LED, buzzer); servo buck -> J3; ESP32 powered over USB (DevKit 5V pin not connected).
Nets connect by name (local labels); every label sits on a pin end
Spec: docs/spec/02-electrical.md §3; board notes: pcb/motion-board/README.md
RackBotGen3 (Delta Cup 2026)
Sheet: /
File: rbg3-motion.kicad_sch
Title: RBG3-MOTION v1: robot motion board
Size: A3
Date: 2026-10-18
Rev: 1
KiCad E.D.A. 10.0.5
Id: 1/1